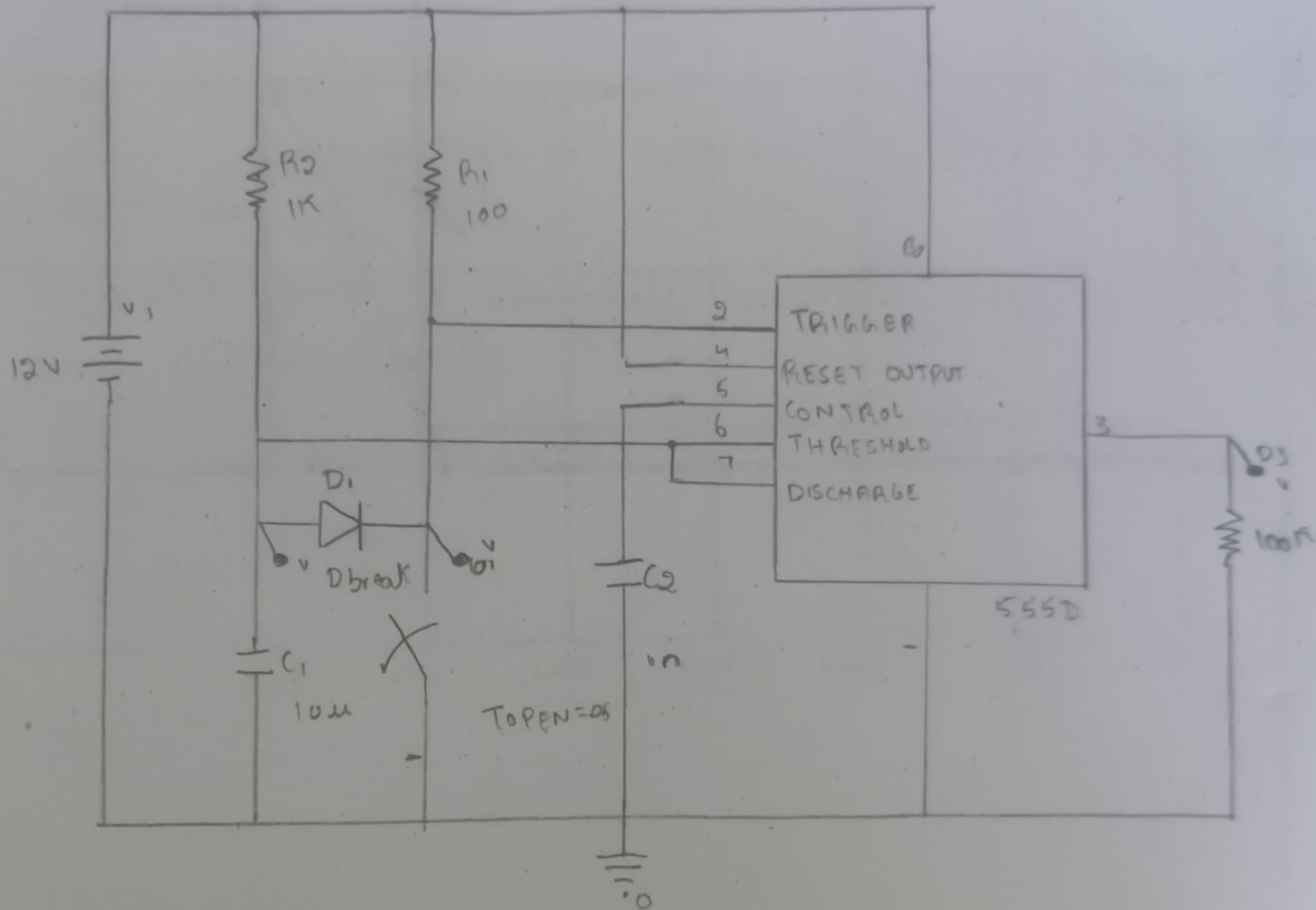


Monostable multivibrator Using 555 Timer

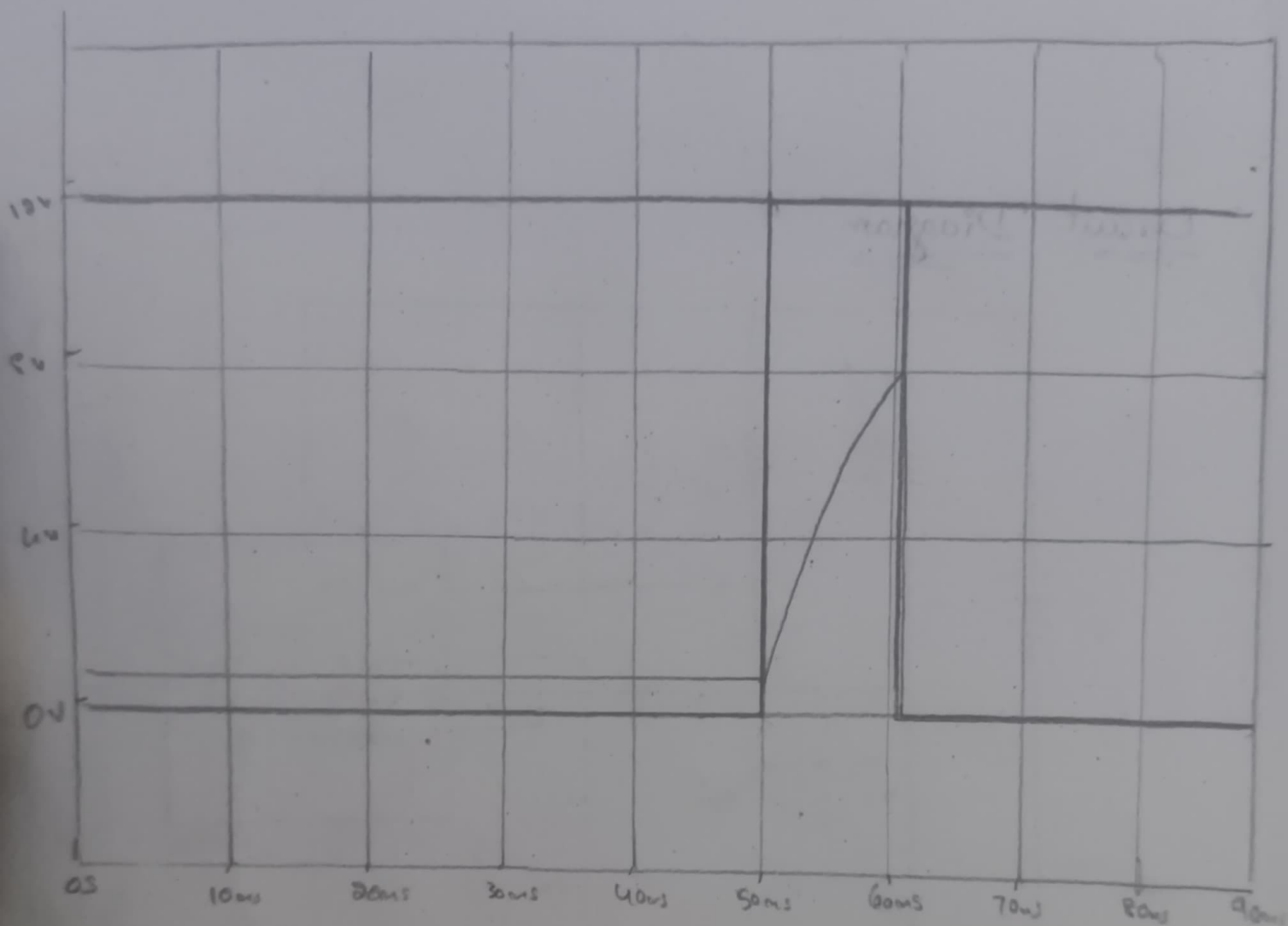
Circuit Diagram



Simulation profile:

Time domain analysis

Run to time = 100ms.

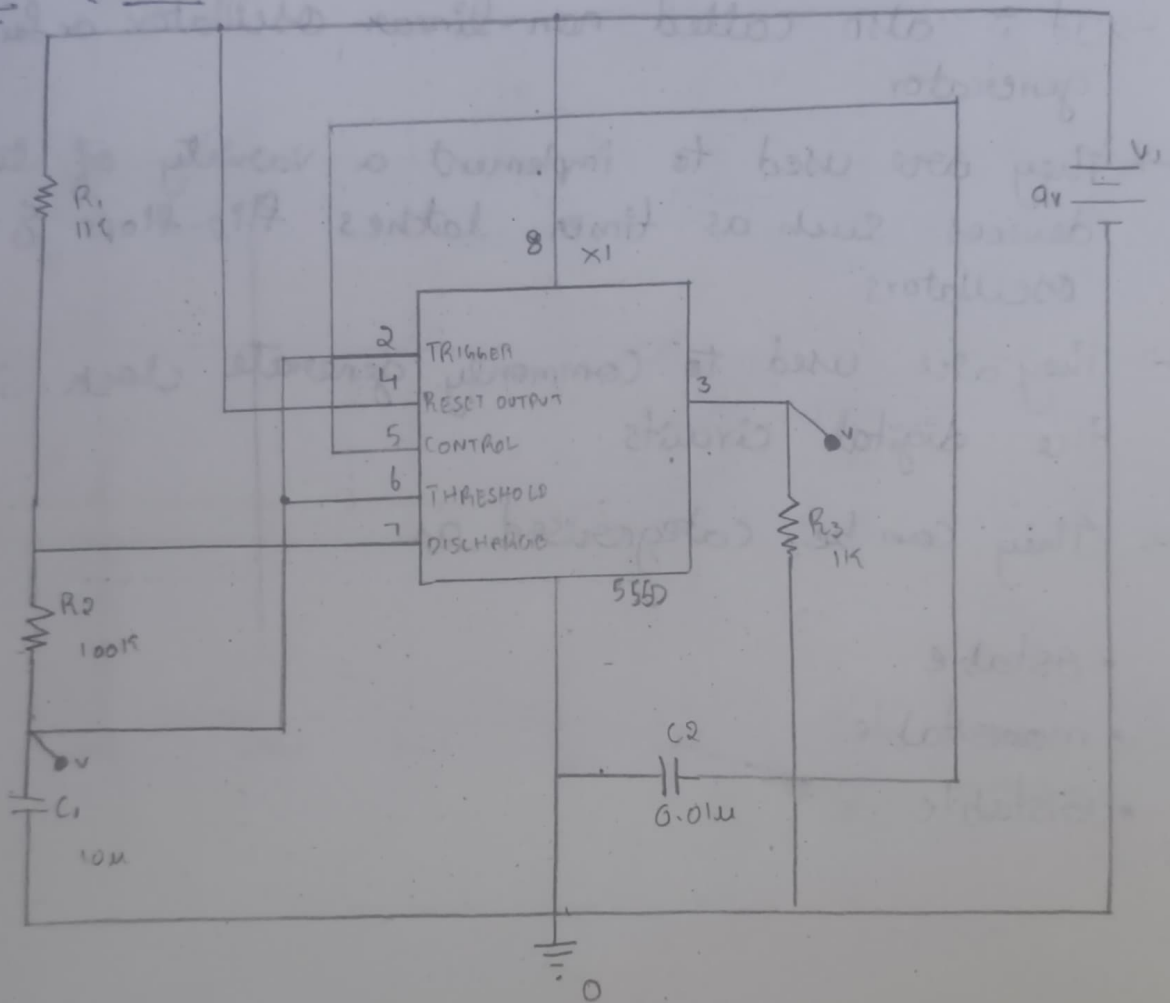


Experiment :- 2

Astable multivibrator using 555 Timer

Pm :- To design and set up a stable multivibrator of IC 555

Circuit diagram :-

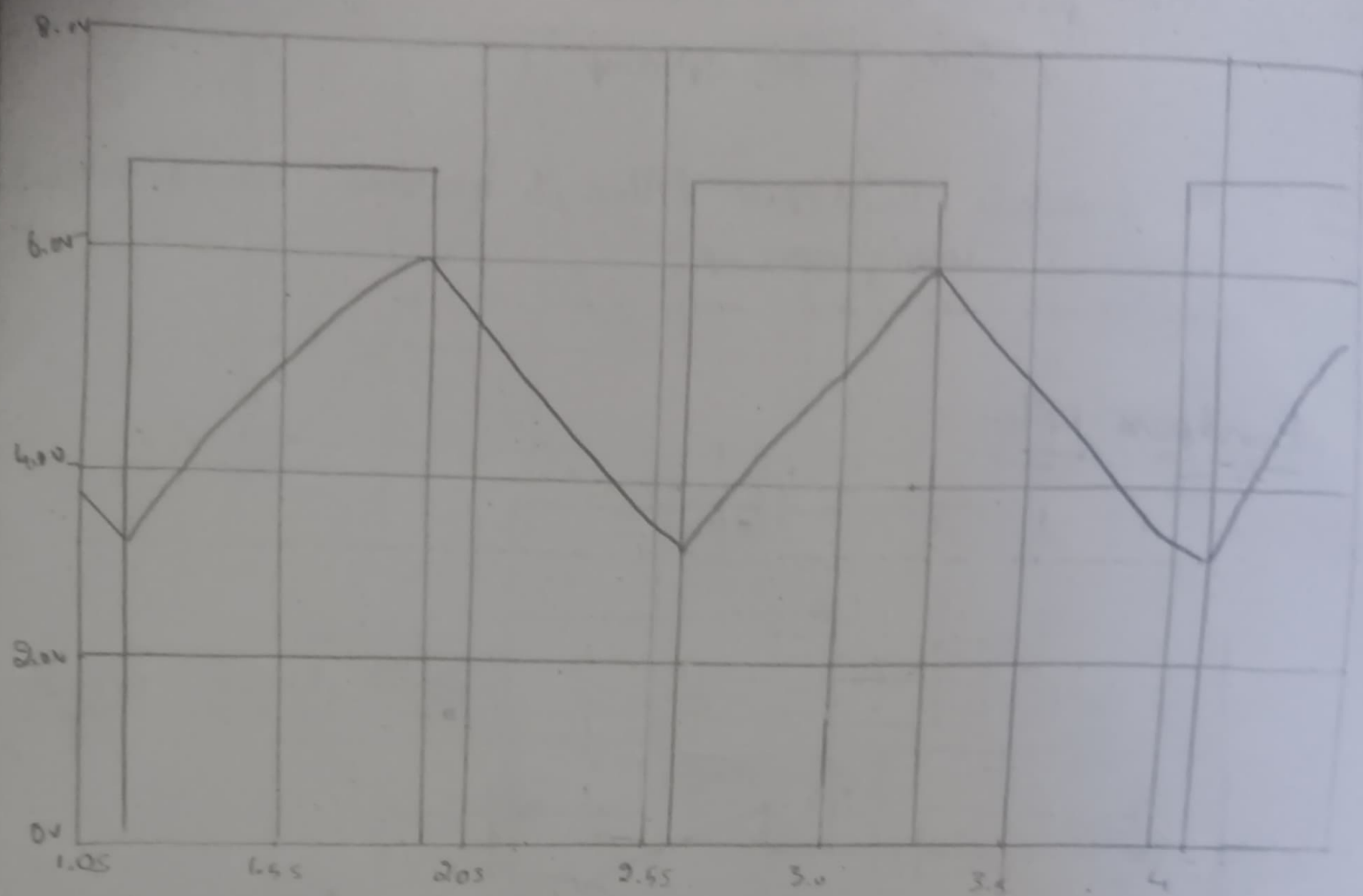


Simulation profile :

Time domain analysis

Run to time = 6s

Start saving data after = 1s

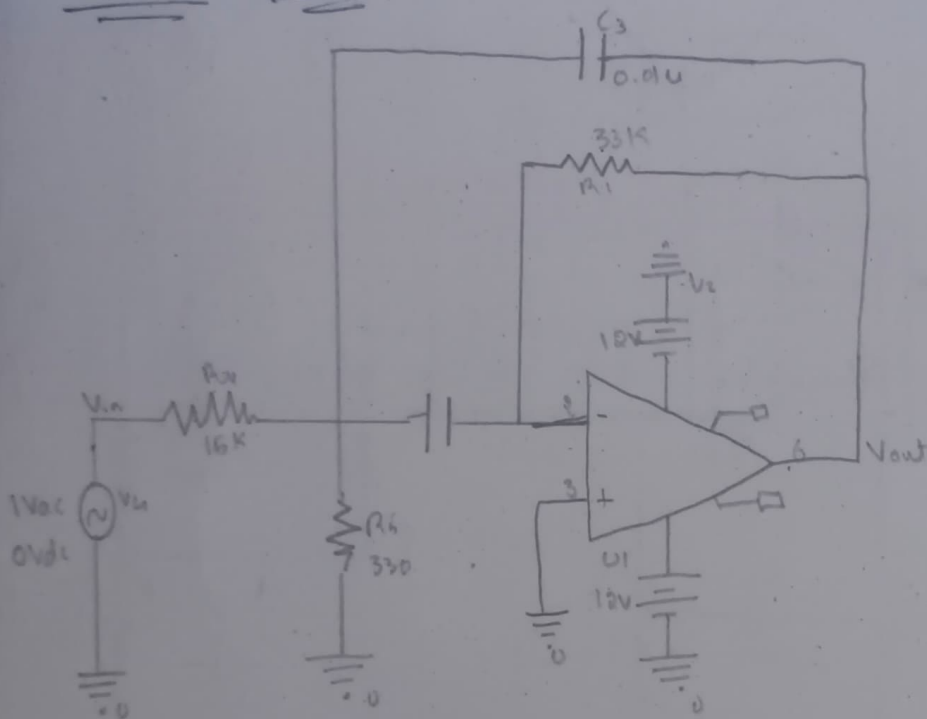


Experiment - 05

Band-pass Filter and Narrow band-reject Filter using op-amp

Aim :- To Simulate Band-pass Filter & Narrow band rejection using op-amp.

Bandpass Filter



Simulation profile

AC Sweep Time

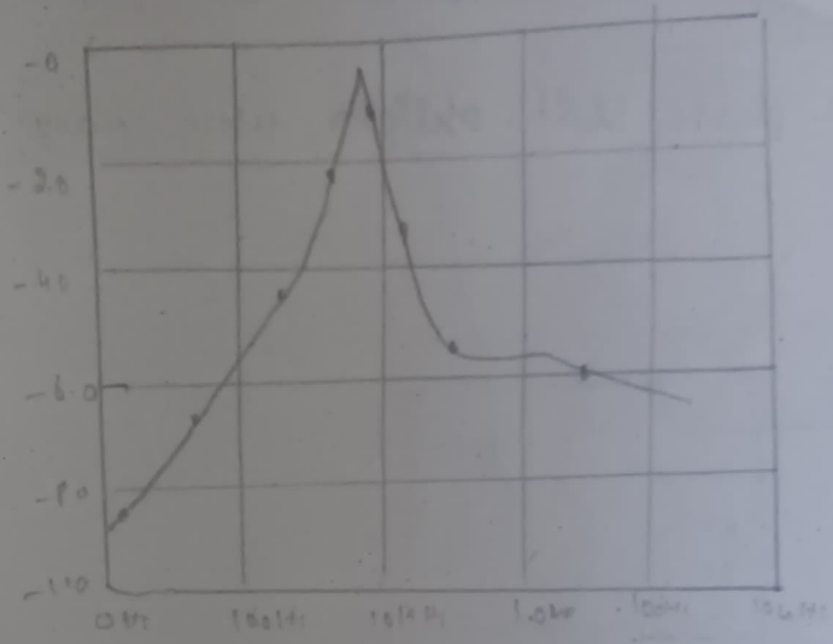
Logarithmic - Decade

Start Frequency = 1

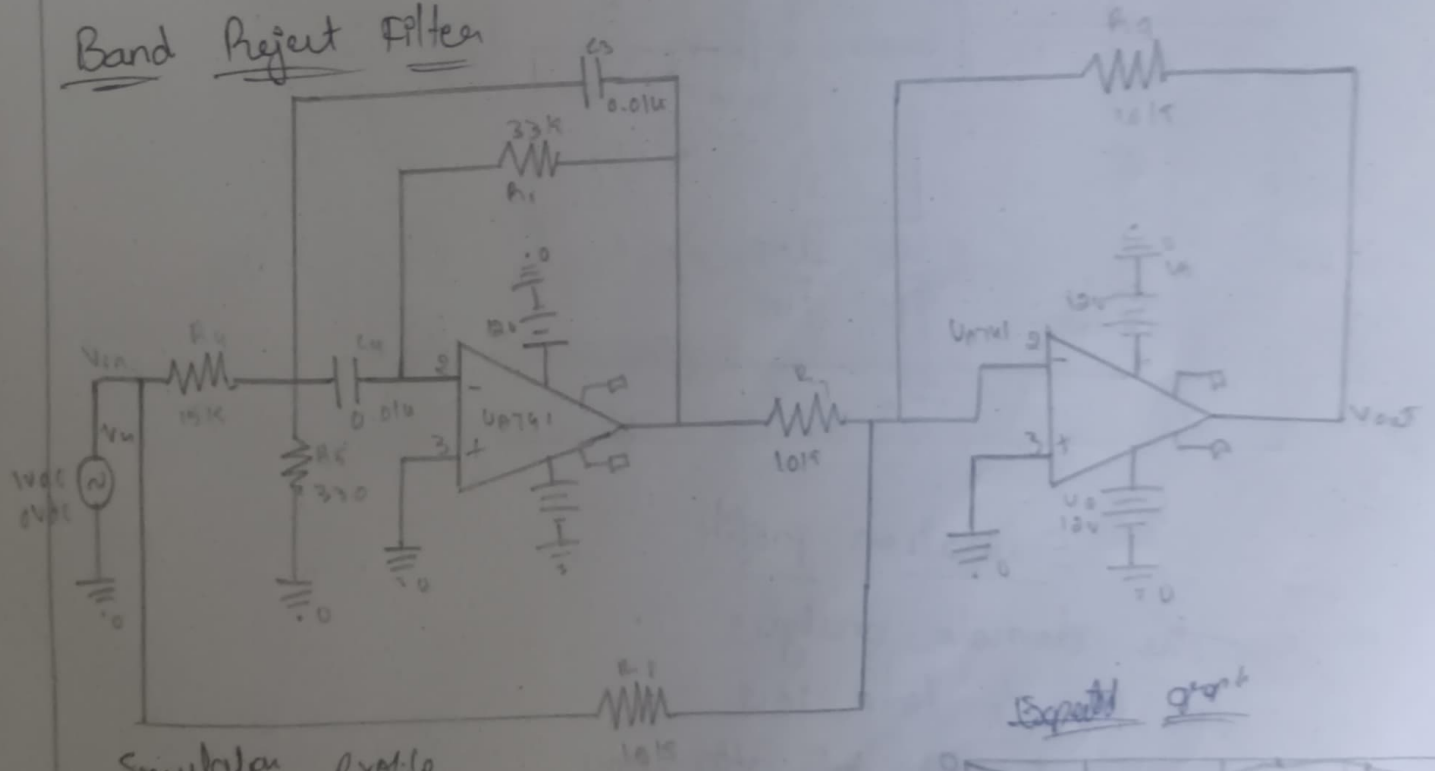
End Frequency = 1k

Points / Decade = 10

Expected Graph



Band Reject Filter

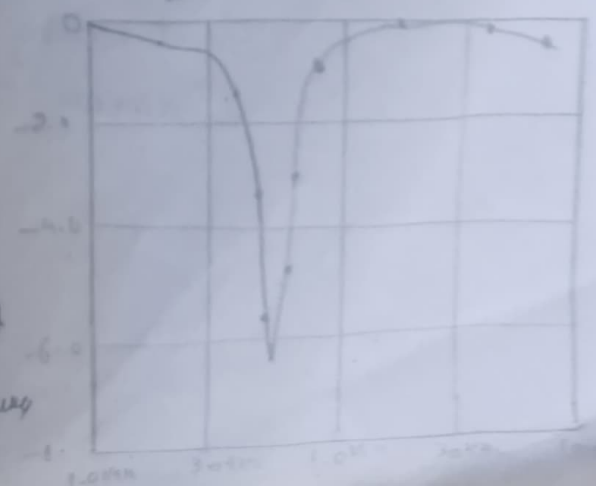


Simulation Profile

Logarithmic Decade
 Start frequency = 1
 End frequency = 100K
 Points / Decade = 10

Result :- Thus the various active filter circuit low pass band pass was designed & its frequency response was analyzed

Expected graph

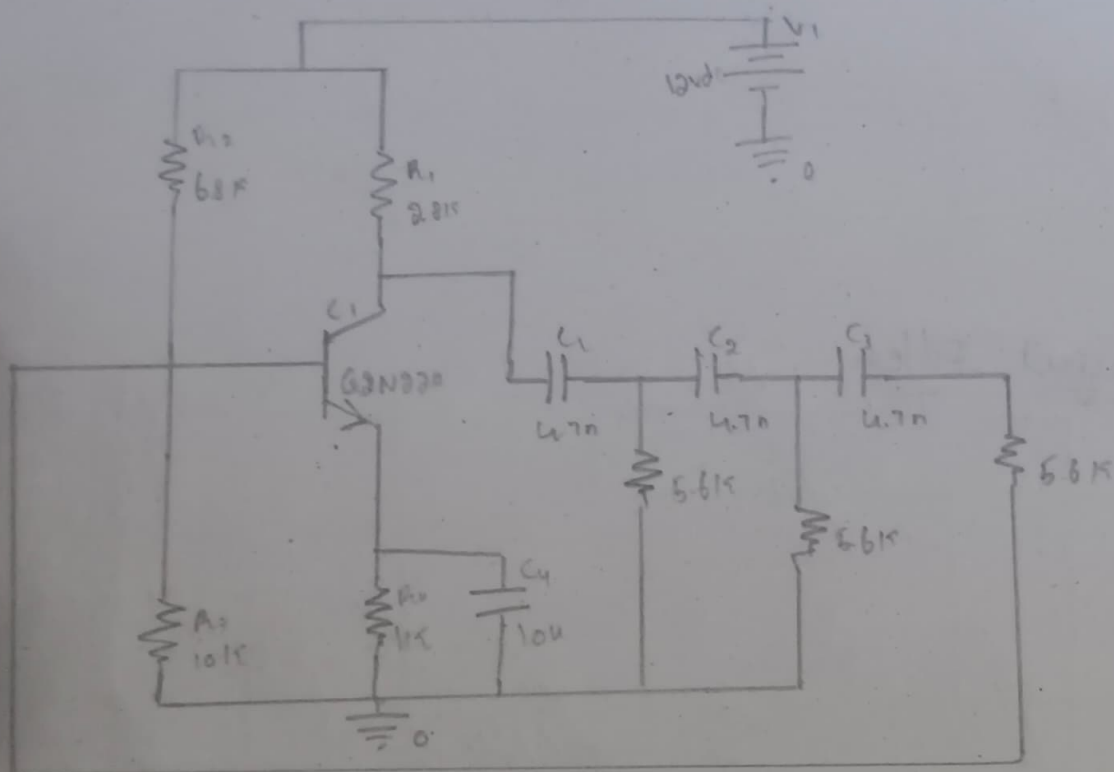


Experiment - 3

Rc phase shift oscillator using op-amp.

Aim:- Simulate Rc phase shift oscillator using op-amp using Pspice

Rc phase shift



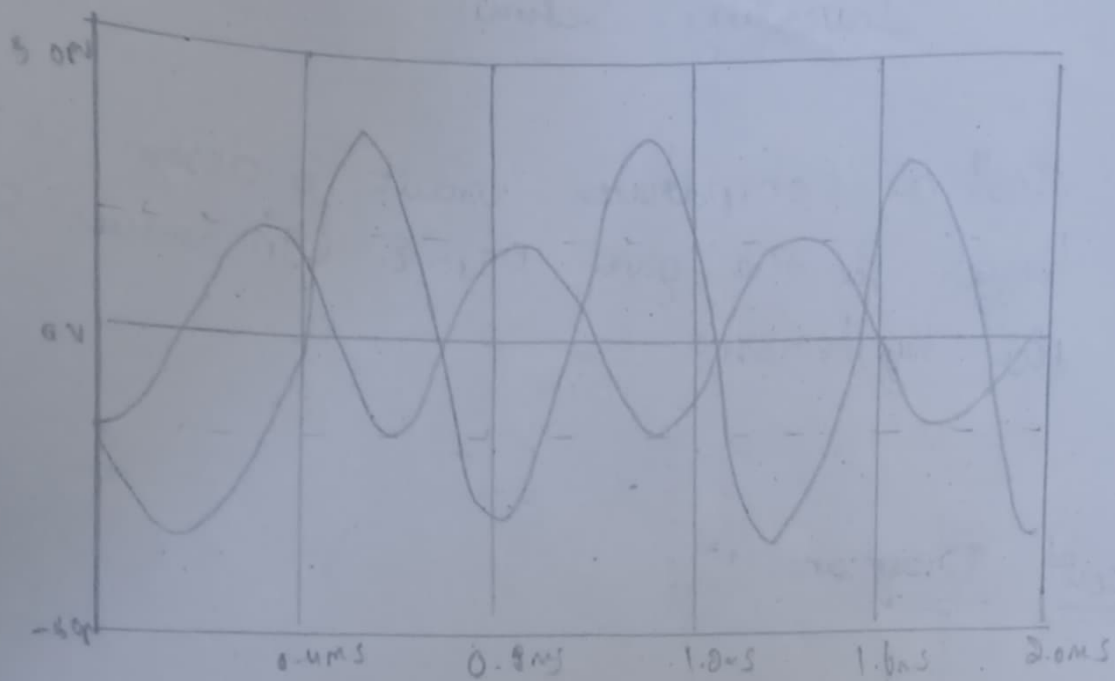
Simulation profile

The domain analysis

Run to time = 1ms

Start saving data after = 0ms

maximum step size = 0



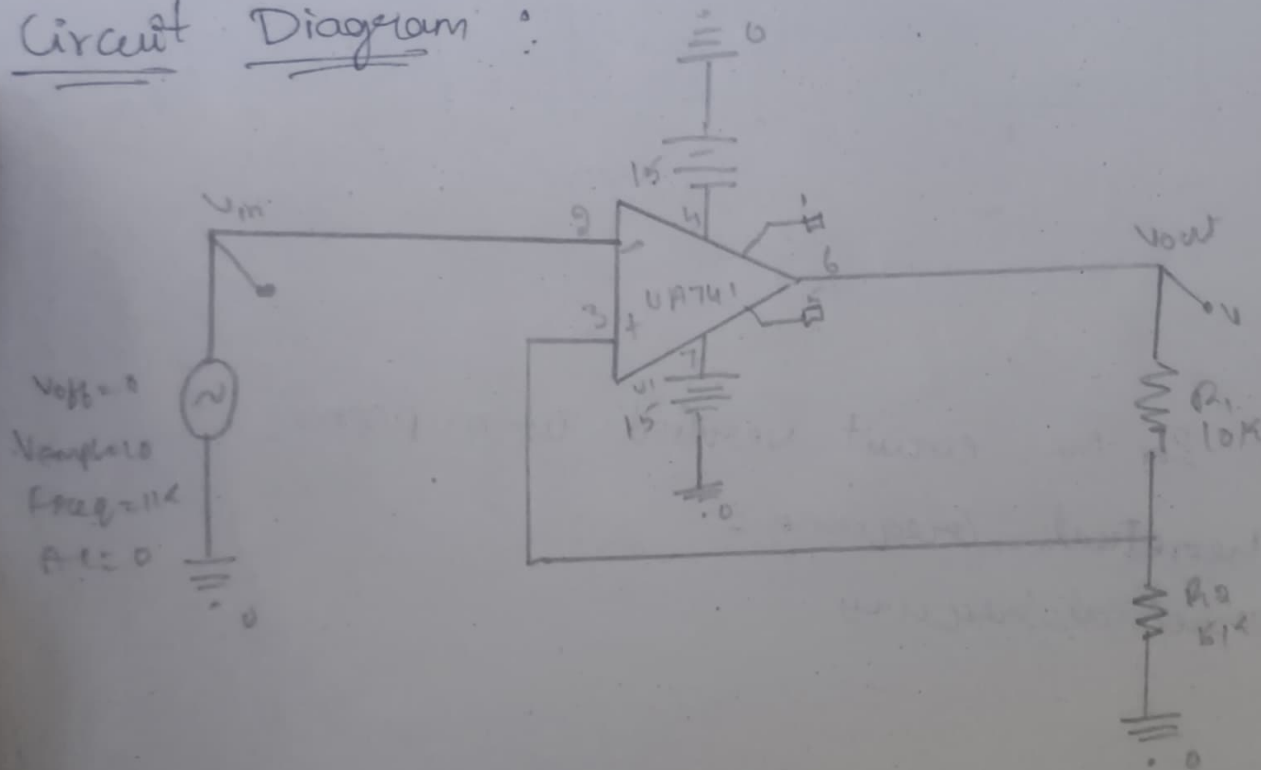
Result.

wave for the circuit verified using PSpice

Theoretical frequency =
practical frequency

Inverting Schmitt Trigger

Aim :- Test a Comparator circuit & design a Schmitt trigger for the given UTP & LTP values & obtain the hysteresis

Circuit Diagram :Simulation profile

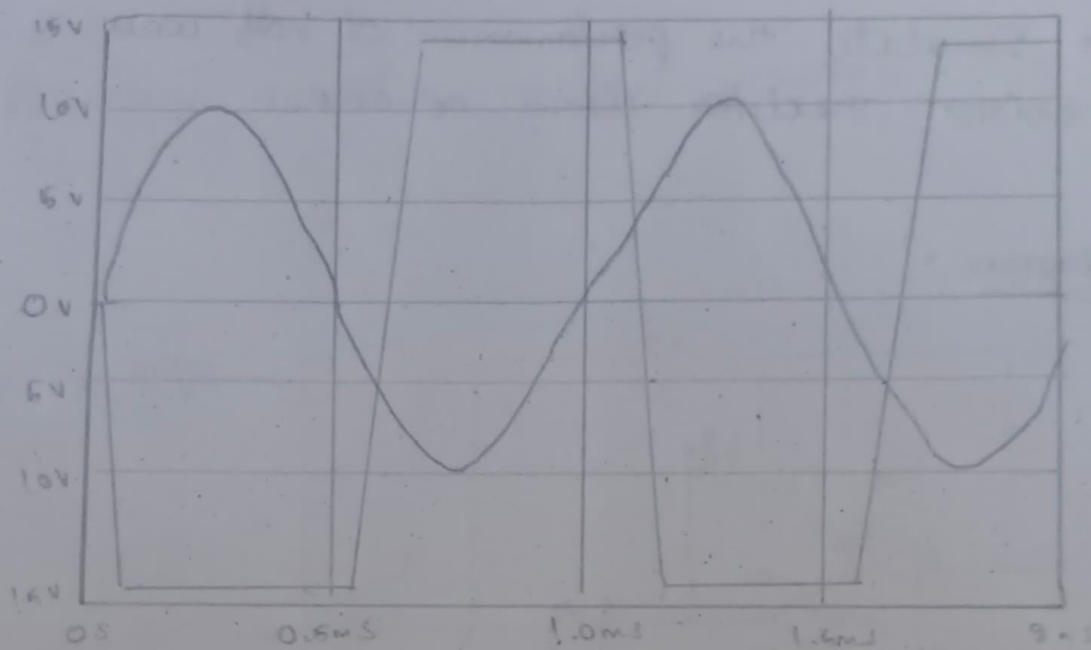
Time domain analysis

Run to time = 2ms

Start saving data after = 0ms

maximum step size = 0

Expected waveform



Result :

The desired UTP & LTP are obtained

$$UTP = +15$$

$$LTP = -15$$

Precision Rectifiers

Aim :- To Simulate the performance of half wave & full wave precision rectifier using op-amp.

Circuit diagram :-

